1. Suppose GDP in 2020 was $21 trillion and GDP in 2021 was $22.5 trillion. What is the growth rate of the economy from 2020 to 2021?

2. If GDP 90 million, Consumption 50 million, Investment 15 million, Government purchases 20 million and exports 8 million, what is the value of imports

3. Consider the following table:

Year	Price of Milk	Quantity of Milk	Price of Honey	Quantity of Honey
2013	$2	100 quarts	$2	50 quarts
2014	$3	200	$4	100
2015	$5	250	$5	200

a. Calculate nominal GDP for 2015:
b. Calculate real GDP for 2015 using 2013 as the base year:
c. Calculate the GDP deflator for 2015 using 2013 as the base year:
d. What is the inflation rate from 2013 to 2014? From 2014 to 2015?

4. The Bureau of Labor Statistics announced in October 2018, of all adult residents of Letterkenny, 3,500 were employed, 500 were unemployed, and 1,000 were not in the labor force. (Note: you will need to calculate the adult population and labor force)

a. Calculate the unemployment rate:
b. Calculate the labor-force participation rate:

5. Consider the following table: *Note that a basket consists of 100 quarts of milk and 50 quarts of honey.*

Year	Price of Milk	Quantity of Milk	Price of Honey	Quantity of Honey
2013	$1	150 quarts	$2	78 quarts
2014	$2	195	$3	102
2015	$3	232	$4	118

a. Using 2013 as a base year, calculate the CPI in 2014:
b. Using 2013 as a base year, calculate the CPI in 2015:
c. Calculate the inflation rate from 2014 to 2015:

6. Suppose your mother received a salary of $45,000 in 1991. What is the value of that salary in 2016 dollars if the CPI in 1991 was 136 and the CPI in 2016 was 240?

7. Real GDP per capita in the United States grew at rate of 2.25 percent from 2010 to 2011. If the United States had continued to grow at this rate how long would it have taken for the economy to double?

8. Consider the following economic information of the economy of Asgard:

- GDP = $500 million
- Consumption = $350 million
- Investment = $100 million
- Government Purchases = $50 million
- Taxes = $75 million
- Transfer Payments = $40 million

Calculate public saving, private saving, and national (aggregate) saving

9. Suppose an economy's GDP has a constant growth rate of 4%. How long will it take for the economy's GDP to double?

10. Draw the graph of the Market for Loanable Funds that demonstrates the effects on interest rate and quantity of investment when there is:

a. crowding out due to government spending or reduction of taxation
b. when government mandates 401K in all firms